

Nayoung Ku

☎ +82 10-8131-0702 ✉ nayoungku1@gmail.com 📄 [nayoung-ku](https://nayoung-ku.github.io) 🌐 [nayoungku1](https://nayoungku1.github.io) 🌐 nayoungku1.github.io

EDUCATION

Handong Global University

B.Sc. in Life Science and Applied Artificial Intelligence (cum laude, GPA: 4/4.5)

North Gyeongsang, South Korea

Feb. 2023 – Feb. 2026

National Institute for Lifelong Education

B.L.A. in Psychology (GPA: 4/4.3)

Seoul, South Korea

Feb. 2021 – Feb. 2023

HONORS & AWARDS

3rd Place, Daewoong Pharmaceutical President's Award (out of 990 teams)

2024

JUMP AI Drug Discovery Competition – Organized by KPBMA & Sponsored by the Ministry of Health and Welfare

Big Data Seagull Scholarship

Spring 2024, Fall 2024

Merit-based scholarship, certified by HGU School of Applied Artificial Intelligence

RESEARCH EXPERIENCE

Undergraduate Researcher → Volunteer Post-baccalaureate Researcher

Aug. 2024 – Present

Handong Global University – Advisor: Prof. Jaeyoung Chun

North Gyeongsang, South Korea

- **Virtual cell modeling (VCC 2026):** Designing a **Mixture-of-Experts (MoE)**, gene regulatory network-aware model architecture to capture **distributional shifts** in gene expression following genetic perturbation, targeting **multiple cell-line generalization (zero-shot)** beyond point-estimate prediction
- Standardizing heterogeneous **Perturb-seq** datasets and pipeline design for a CRISPRi-based gene expression perturbation prediction model submitted toward the **Arc Institute Virtual Cell Challenge (VCC) 2026**
- **Single-Cell Demultiplexing:** Developed a diffusion-based probabilistic demultiplexing pipeline for hashed scRNA-seq data that learns the underlying signal distribution rather than relying on hard-threshold cutoffs
- **Circulating Tumor Cell (CTC) Detection:** Constructed a high-quality annotated training dataset via Roboflow for real-time CTCs detection in unstained blood **microscopy images** and conducted a systematic literature review on CTC morphology to ensure biological accuracy across heterogeneous cell populations
- **Drug Potency Prediction (IC50 Regression):** Initiated and led an **IC50 regression** project predicting drug potency from SMILES strings, engineering molecular descriptors and designing a custom feature selection method for traditional ML regression models
- Achieved **top-3 finalist** status (out of 990 teams) at a national AI drug discovery challenge, receiving the Daewoong Pharmaceutical President's Award (see Honors & Awards)

Visiting Undergraduate Researcher

Jun. 2025 – Jul. 2025

University of Nevada, Las Vegas – Advisor: Prof. Mingon Kang (DataX Lab)

Nevada, United States

- Developed a lightweight **patch tiling and filtering algorithm** for **whole slide images (WSIs)** from the **TCIA CMB-LCA** histopathology dataset, reducing raw data volume by **80%** (128GB → 25GB) without sacrificing model-critical features, significantly lowering computational overhead
- Built a hybrid deep learning framework integrating **Vision Transformer** (Prov-GigaPath) embeddings with an **LSTM**-based classifier, achieving **AUC 0.99** on large-scale, imbalanced tumor classification across WSIs

Undergraduate Researcher

Jan. 2025 – Apr. 2025

Seoul National University – Advisor: Prof. Dr. Howard Lee (CCADD)

Seoul, South Korea

- Contributed to a **NLP research project commissioned by Korea's Ministry of Food and Drug Safety (MFDS)**, building a **Korean-English parallel sentence corpus** of pharmaceutical regulatory documents
- Standardized data collection, annotation, and management protocols across a multi-source document corpus, establishing a reusable curation pipeline for the research team, and designed a sentence alignment algorithm for automated parallel corpus construction from heterogeneous document sources

Undergraduate Research Intern

Mar. 2024 – Nov. 2024

Handong Global University – Advisor: Prof. Taejin Ahn (BioData Lab)

North Gyeongsang, South Korea

- Analyzed clinical **Tumor-Educated Platelet (TEP) RNA-seq** datasets to identify **ovarian and colorectal cancer biomarkers**, applying feature selection and hierarchical clustering to isolate signal from high data sparsity

- Developed a **transformer-based preprocessing framework** encoding raw RNA sequences of TEPs directly into embeddings via **ProtBERT**, bypassing traditional count-matrix construction, as an independent capstone project and achieved **AUC 0.71** in ovarian cancer classification and demonstrating that embeddings capture latent biological structure without manual feature engineering

PRESENTATIONS

Conference & Capstone Presentations

No Cell Left Behind: Recovering Discarded Cells via Self-Supervised Diffusion Denoising	Dec. 2025
<i>Handong Global University Bio-AI Capstone Design Fair</i>	<i>Oral Presentation</i>
Diagnosing Cancer with Genomic Context	Nov. 2024
<i>Handong Global University Capstone Design in Life Science Fair</i>	<i>Oral Presentation</i>
IRAK4 IC50 Inhibition Prediction	Oct. 2024
<i>AI Pharma Korea Conference 2024</i>	<i>Poster Presentation</i>

TECHNICAL SKILLS

Languages: Python, R, SQL, Bash
ML/DL: PyTorch, scikit-learn, Weights & Biases, MLflow, Optuna, HuggingFace
Bioinformatics: Scanpy, scRNA-seq Analysis, Bulk RNA-seq, Histopathology & Microscopy Image Analysis
Infrastructure: Git, GitHub, Slurm, LSF, MIG-partitioned GPU clusters, Linux (Ubuntu, ARM)

TEACHING EXPERIENCE

Introduction to AI and Machine Learning, <i>Teaching Assistant</i>	Aug. 2025
<i>Addis Ababa Science and Technology University – Instructor: Prof. Dai Yon Cho</i>	<i>Addis Ababa, Ethiopia</i>
Big Data Modeling and Platforms, <i>Teaching Assistant</i>	Mar. 2025 – Jun. 2025
<i>Handong Global University – Instructor: Prof. Jaeyoung Chun</i>	<i>Pohang, South Korea</i>
AI Application Programming, <i>Teaching Assistant</i>	Mar 2025. – Jun. 2025
<i>Handong Global University – Instructor: Prof. Sangsan Lee</i>	<i>Pohang, South Korea</i>

COMMUNITY INVOLVEMENT

R Programming Camp, <i>Teaching Assistant</i>	Jan 2026
<i>Mentored high school students in data analysis and visualization using R</i>	<i>Pohang, South Korea</i>
Handong Office of International Affairs, <i>Exchange Student Mentor</i>	Aug – Dec 2024
<i>Assisted exchange students with adaptation to Korean culture and residential college</i>	<i>Pohang, South Korea</i>
Incheon Beauty and Art High School Python Camp, <i>Teaching Assistant</i>	Aug 2023
<i>Supported Python programming instruction for elementary to high school students</i>	<i>Incheon, South Korea</i>